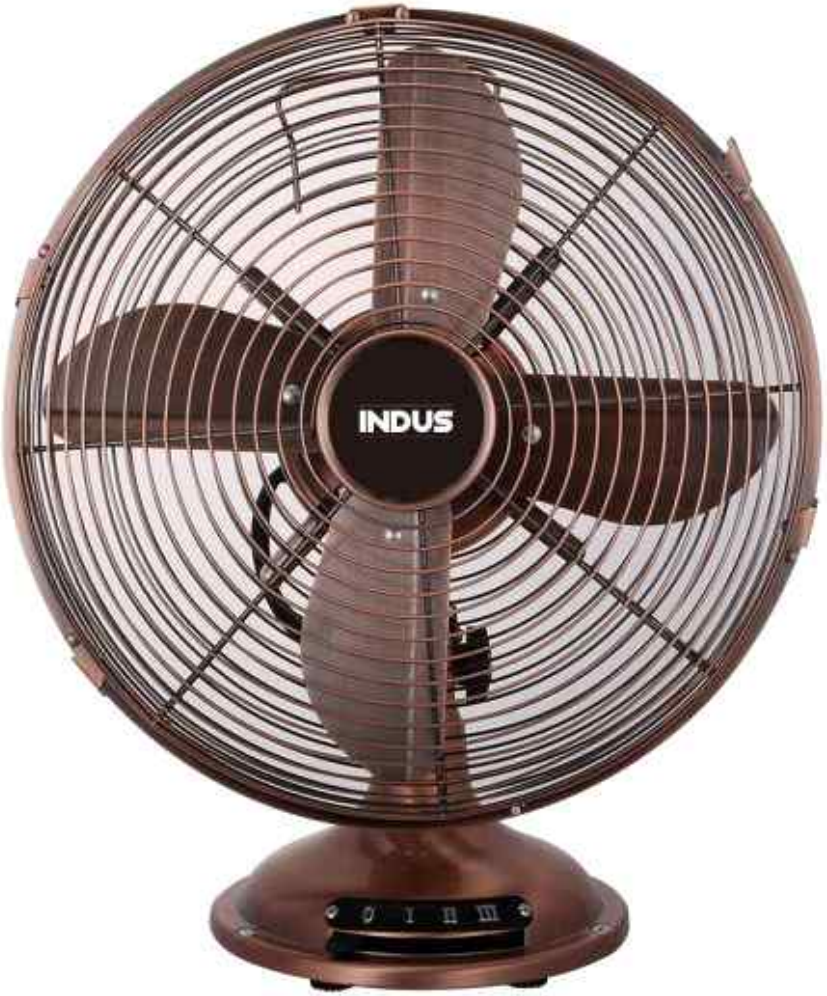
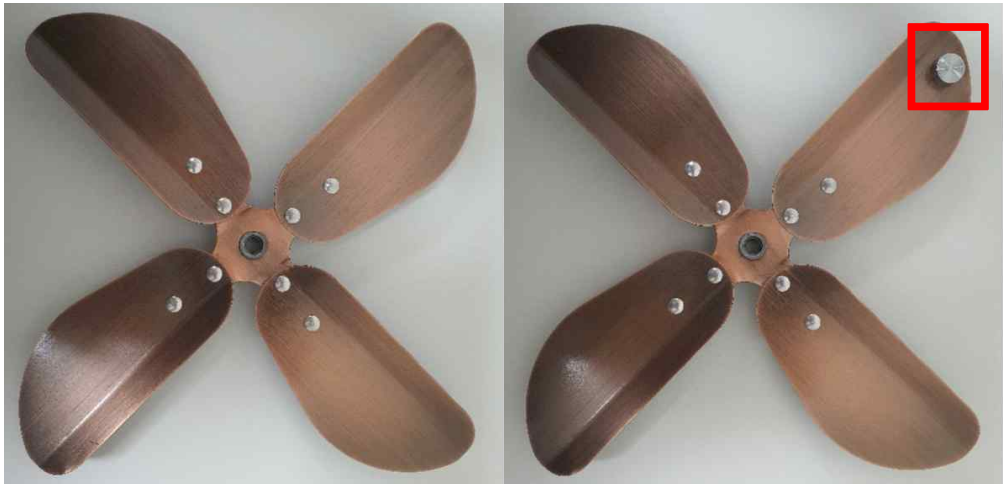
진단 실습

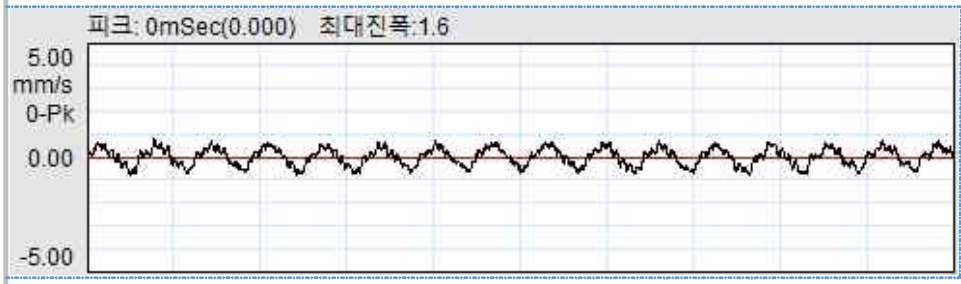
https://cafe.daum.net/VibrationAnalysis

실습1_FAN 불 평형 1x 진동 진단 실습

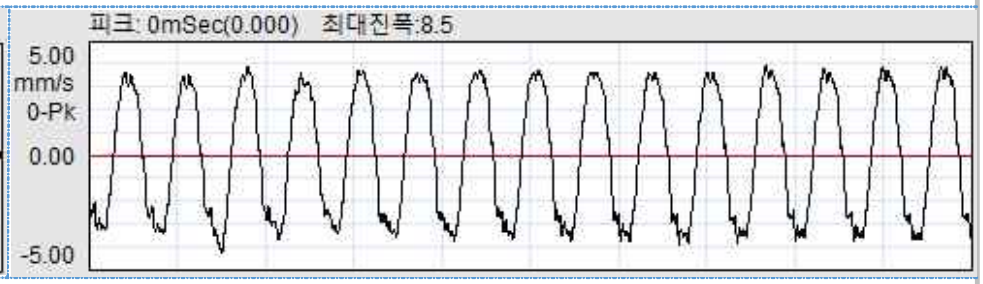




1x 회전 성분 불 평형 시간파형 진단 보고서

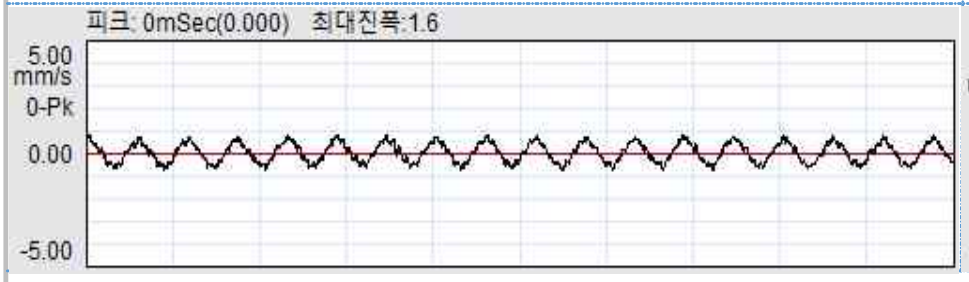


정상

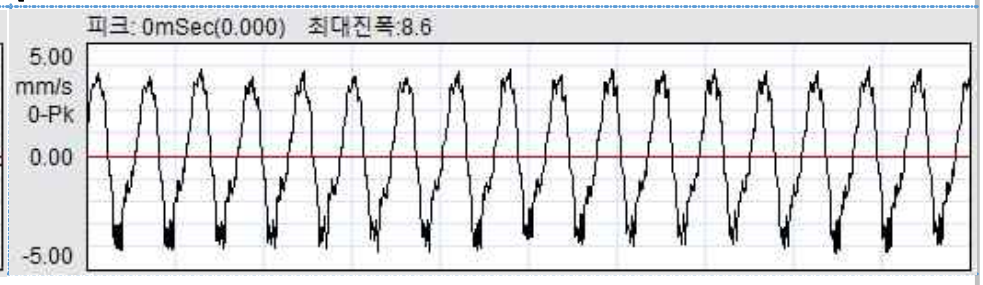


무개 추 장착

[그림 1] 1단

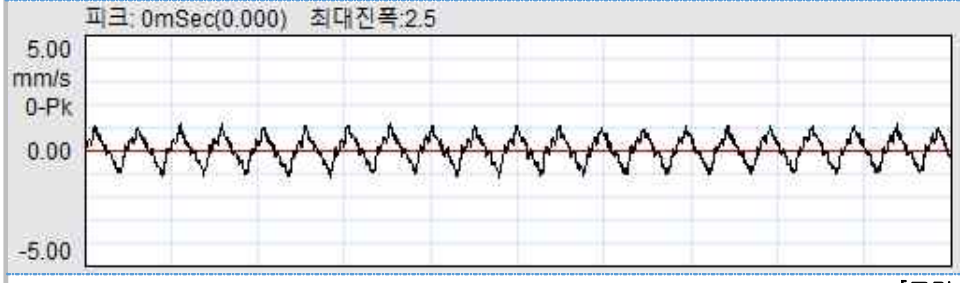


정상

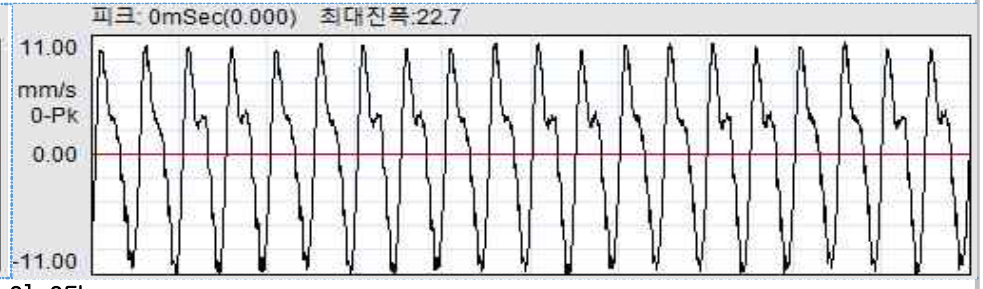


무개 추 장착

[그림 2] 2단



정상



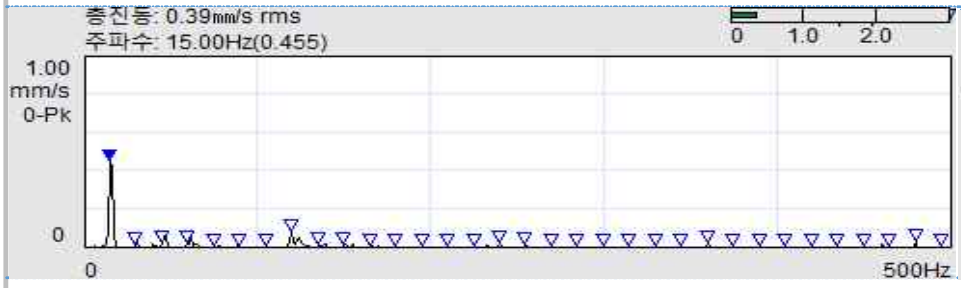
무개 추 장착

[그림 3] 3단

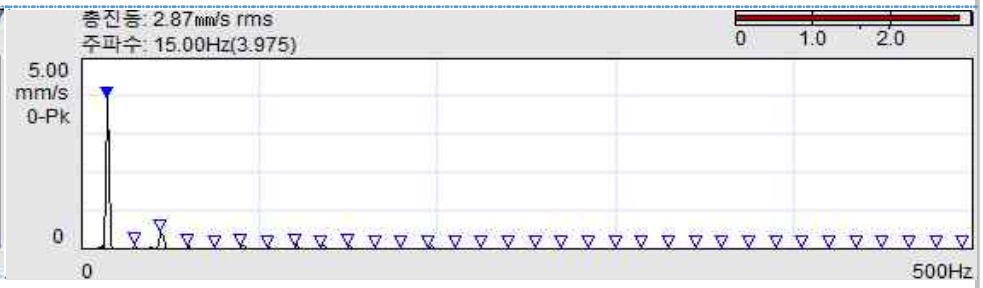
◆ FFT Spectrum

- 회전 속도 별 정상/비정상 진폭 값?
- 회전 속도 별 시간 파형 주기?
- 진동의 원인?

1x 회전 성분 불 평형 스펙트럼 진단 보고서

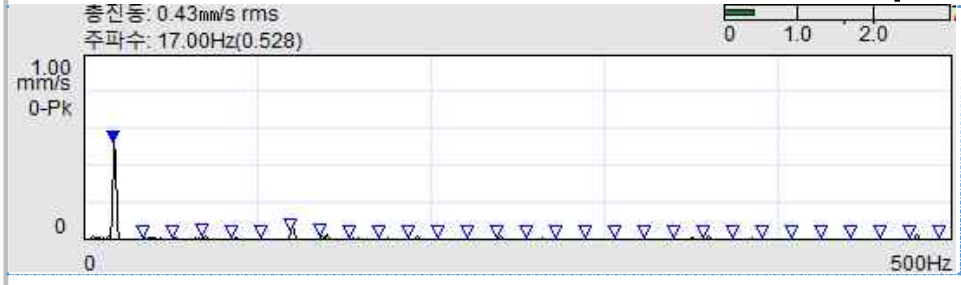


정상

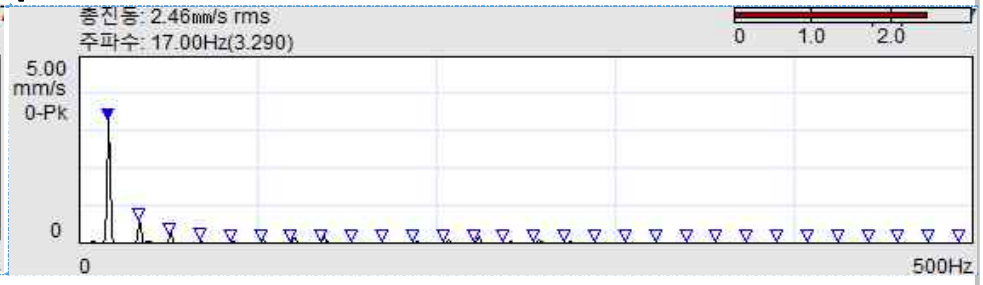


무개 추 장착

[그림 1] 1단

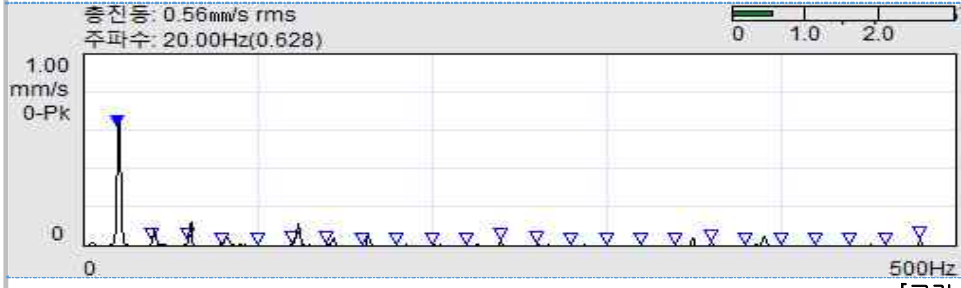


정상

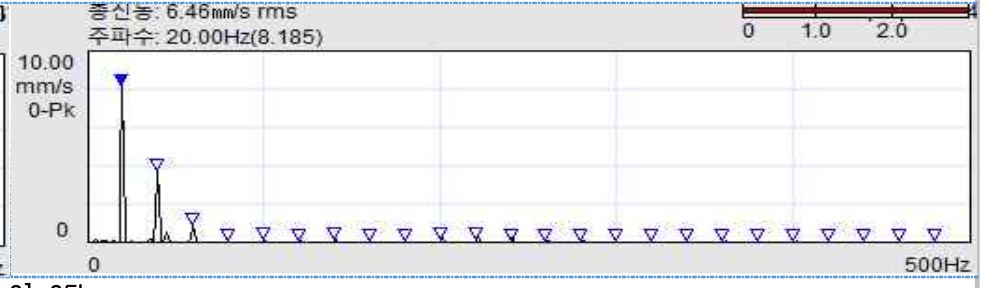


무개 추 장착

[그림 2] 2단



정상



무개 추 장착

[그림 3] 3단

◆ FFT Spectrum

- 정상/비정상 속도별 1x Peak값?
- 1x Peak값이 가장 큰 단수?
- 진동의 원인?

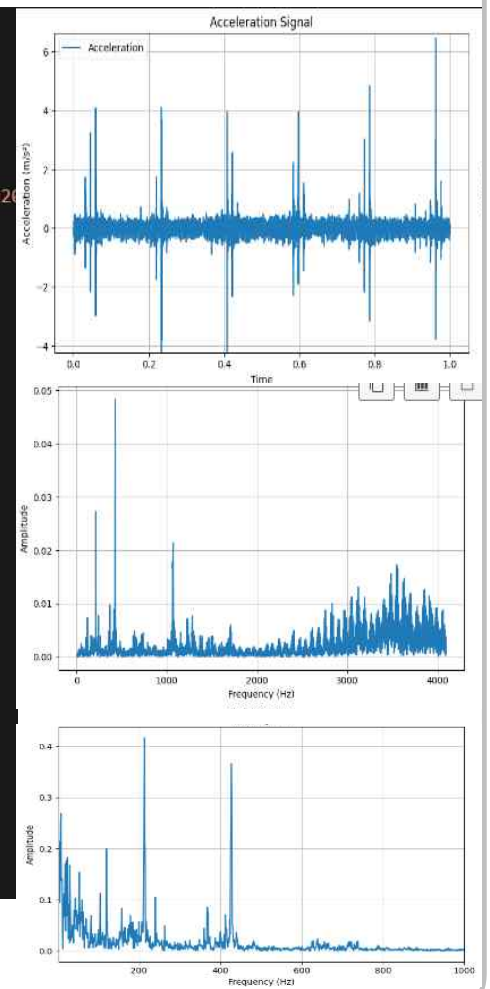
실습2-신호 처리 실습(조별 결함Raw Data 신호처리)

예제)

A_ 2조 Bearing 볼 결함

X	Series0
0	-0.10986328125
0.0001220703125	0.1434326171875
0.000244140625	0.10223388671875
0.0003662109375	0.13580322265625
0.00048828125	0.12664794921875
0.0006103515625	0.14801025390625
0.000732421875	0.1617431640625
0.0008544921875	0.28076171875
0.0009765625	0.384521484375
0.0010986328125	0.30670166015625
0.001220703125	0.372314453125
0.0013427734375	0.1983642578125
0.00146484375	0.1617431640625
0.0015869140625	0.03509521484375
0.001708984375	0.054931640625
0.0018310546875	0.03204345703125

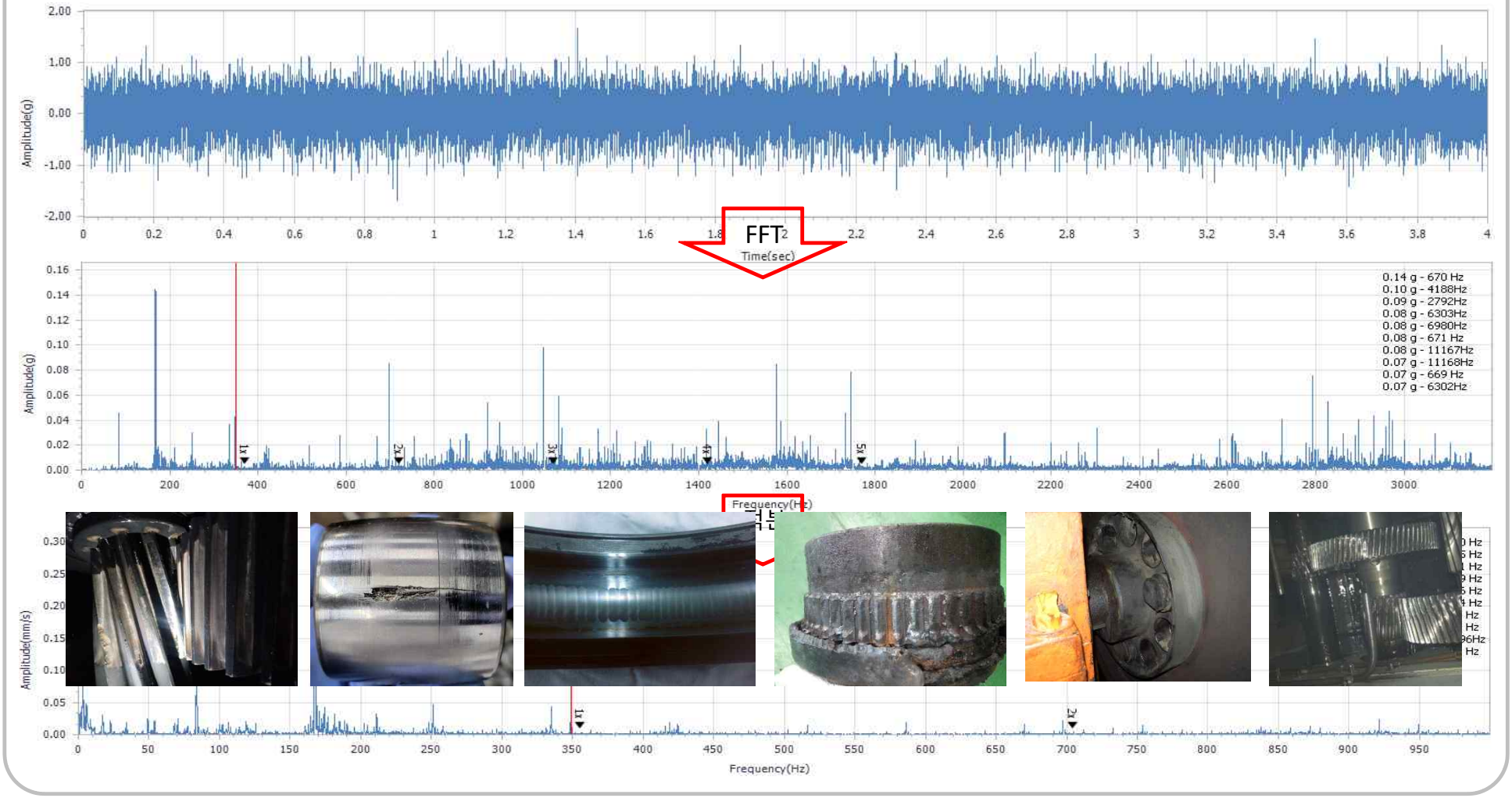
```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy import integrate

# CSV 파일 읽기
df = pd.read_csv('C:/Users/sam17/AppData/Local/Packages/CanonicalGroupLimited.Ubuntu2

time = df.iloc[:, 0]
acceleration = df.iloc[:, 1]

y_min = min(acceleration)
y_max = max(acceleration)

# FFT 수행
fft_result = np.fft.fft(acceleration)
n = len(time)
freq = np.fft.fftfreq(n, d=(time[1]-time[0]))

fft_magnitude = np.abs(fft_result) / n

# 2Hz 이하 주파수 필터링
cutoff_freq = 2
fft_filtered = fft_result.copy()
fft_filtered[np.abs(freq) <= cutoff_freq] = 0

# 필터링된 가속도 신호
filtered_acceleration = np.real(np.fft.ifft(fft_filtered))

# 속도와 변위 계산
conversion_factor = 39.3701
velocity = integrate.cumulative_trapezoid(filtered_acceleration, time, initial=0)
```



신호 처리 알고리즘 프로젝트 실습

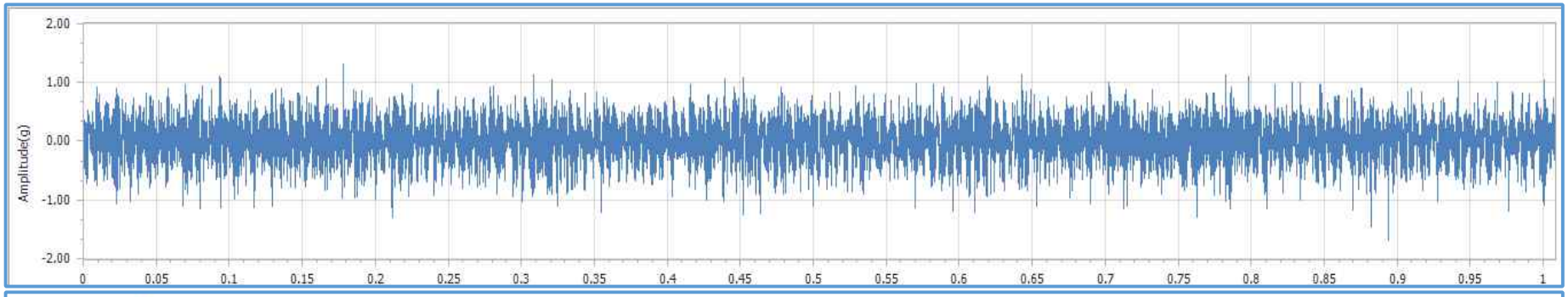
1. Raw Data > Time waveform 변환 > 가속도 Spectrum > 속도 Spectrum 변환

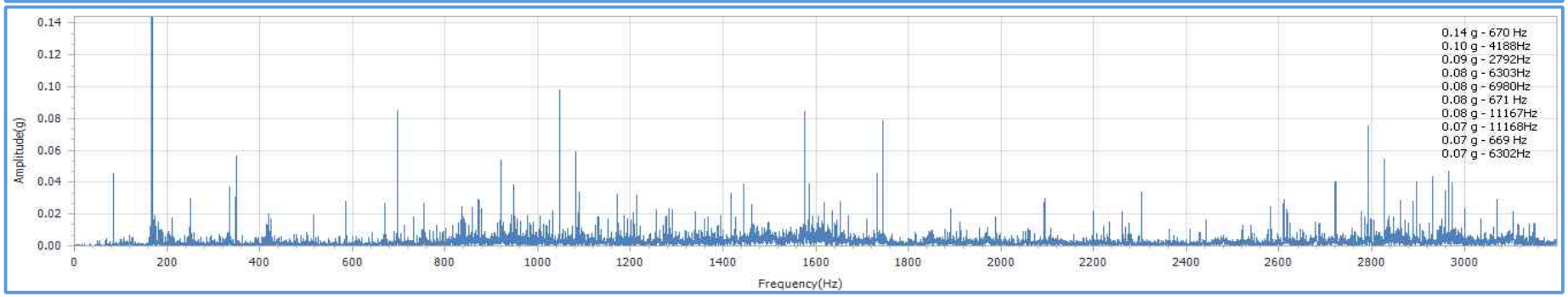
- 1조 Gear/Bearing
- 2조 볼 Bearing
- 3조 외륜 Bearing
- 4조 Coupling
- 5조 Gear GMF



신호 처리 프로젝트 실습 1조

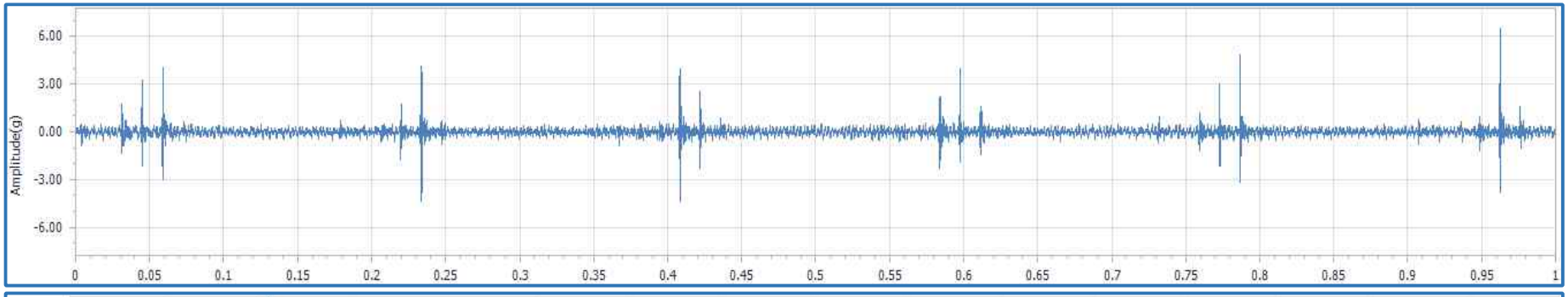
▶ 1조 압출기 감속기
Gear/Bearing 손상 사례

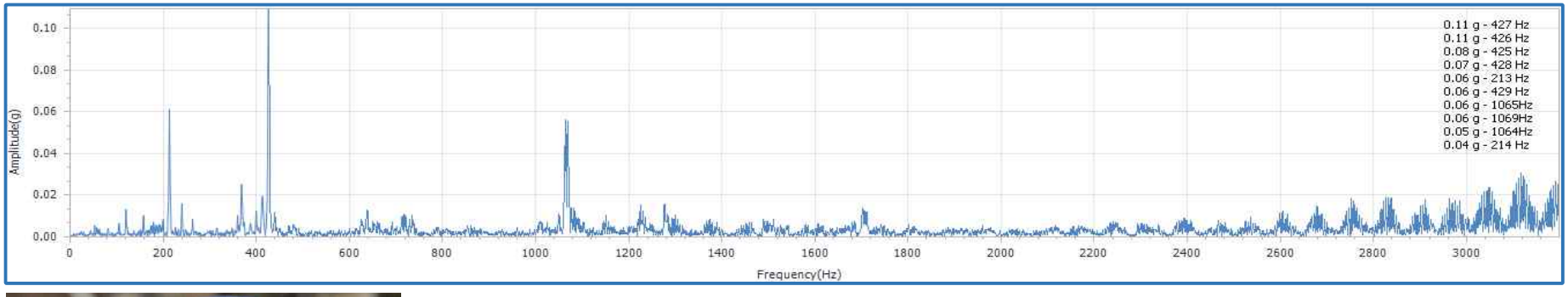




신호 처리 프로젝트 실습 2조

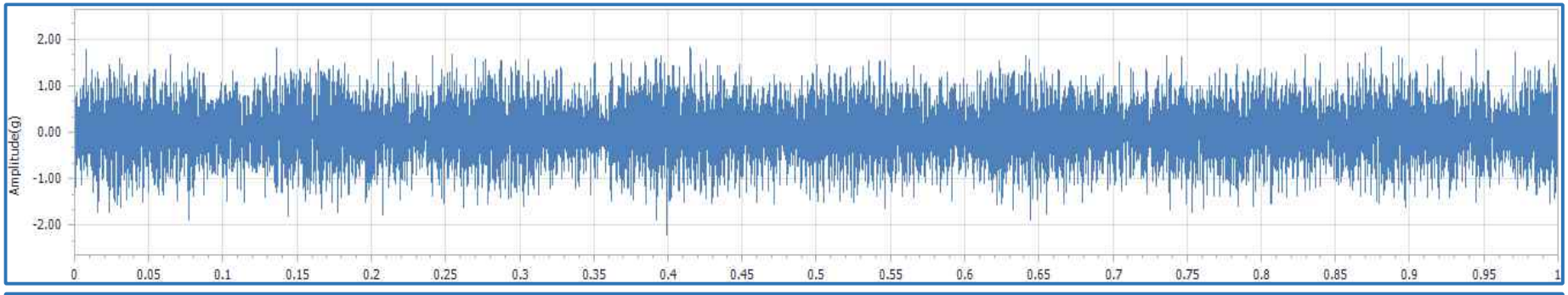
▶ 2조 Bearing Ball 결함

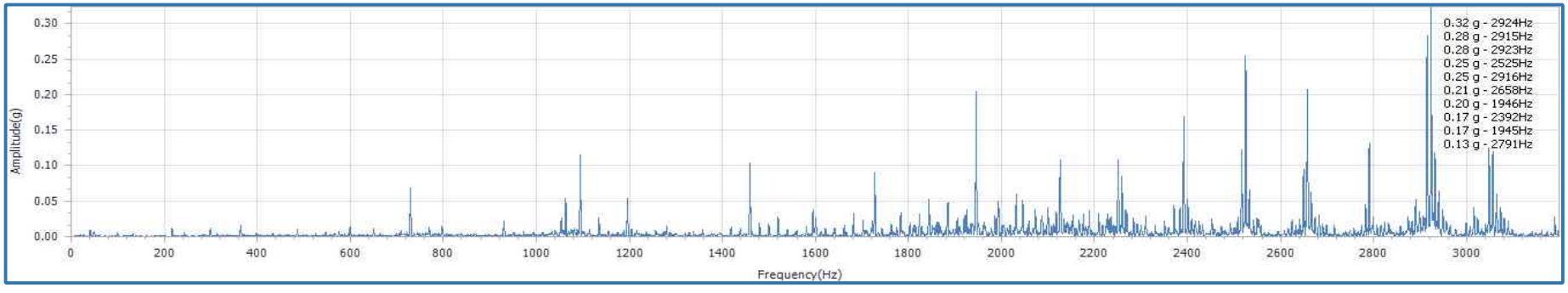




신호 처리 프로젝트 실습 3조

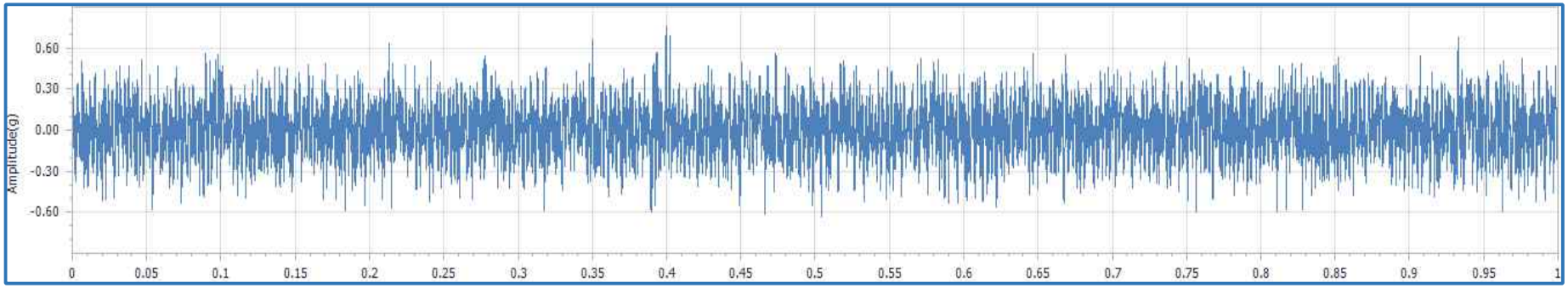
▶ 3조 Bearing 외륜 결함

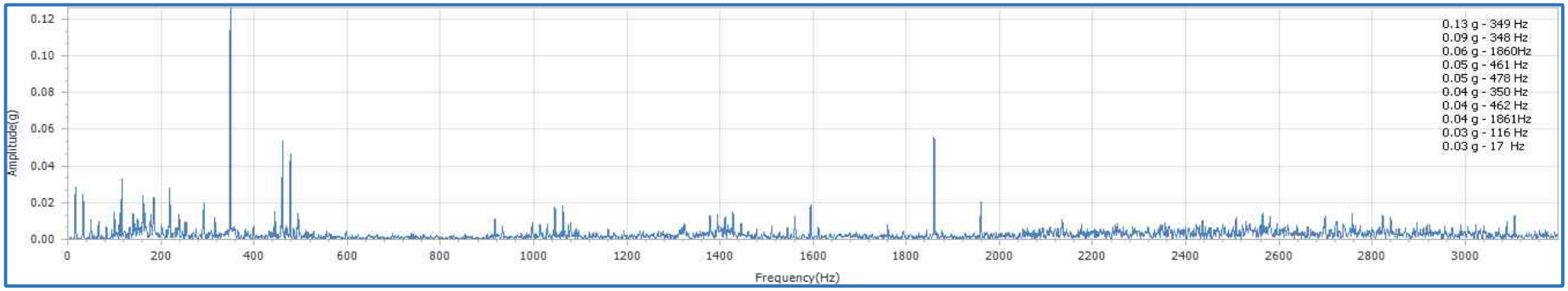




신호 처리 프로젝트 실습 4조

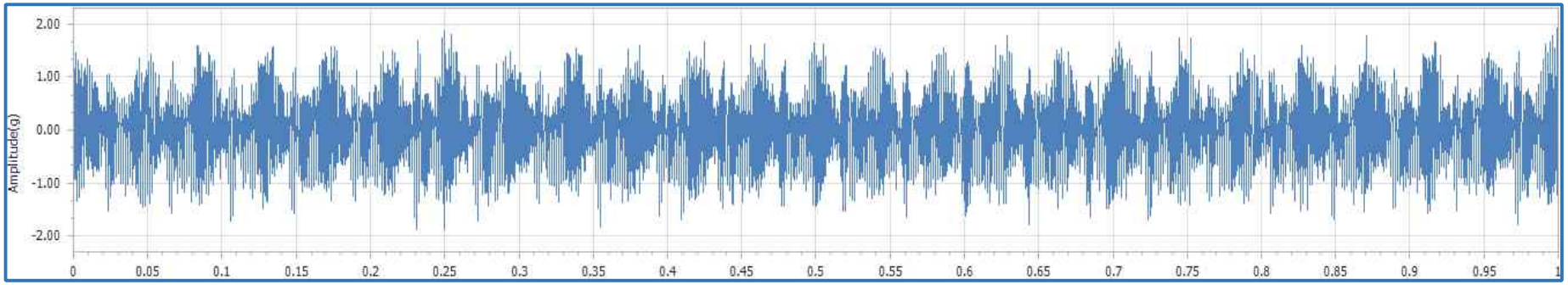
▶ 4조 Coupling 결함

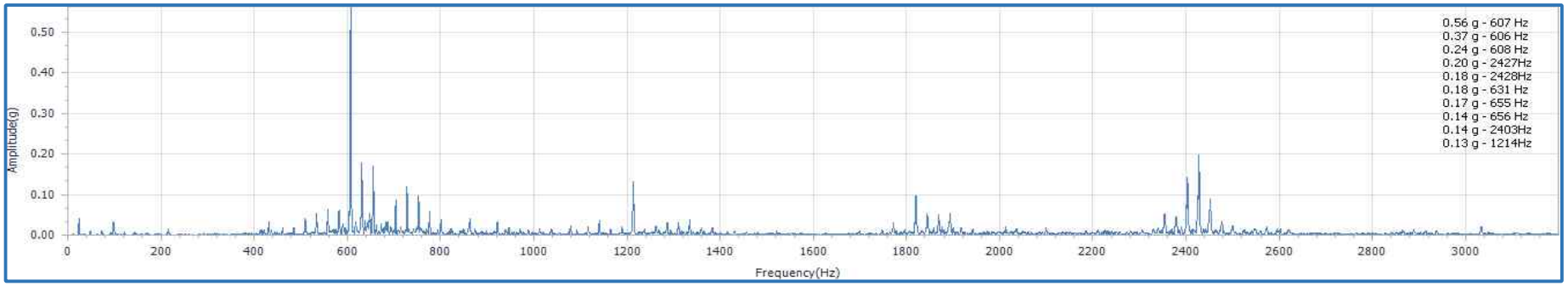




신호 처리 프로젝트 실습 5조

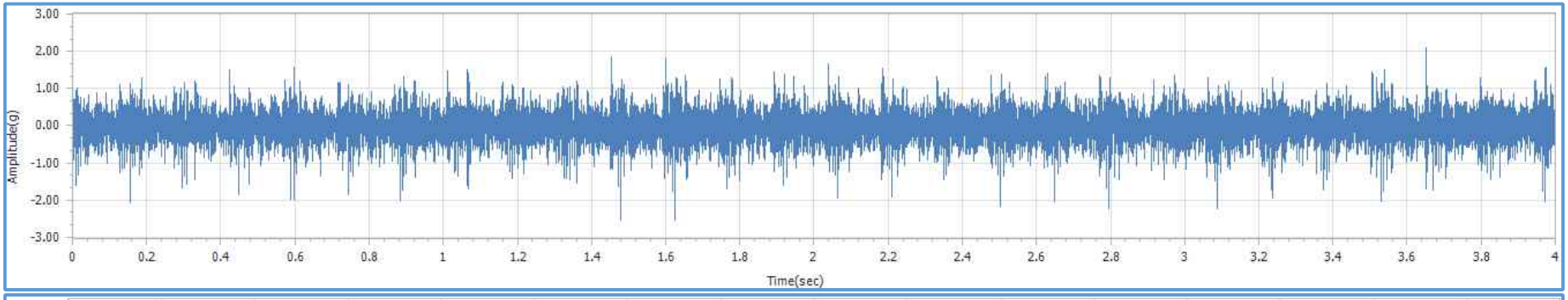
▶ 5조 감속기 GMF 커플링 마모

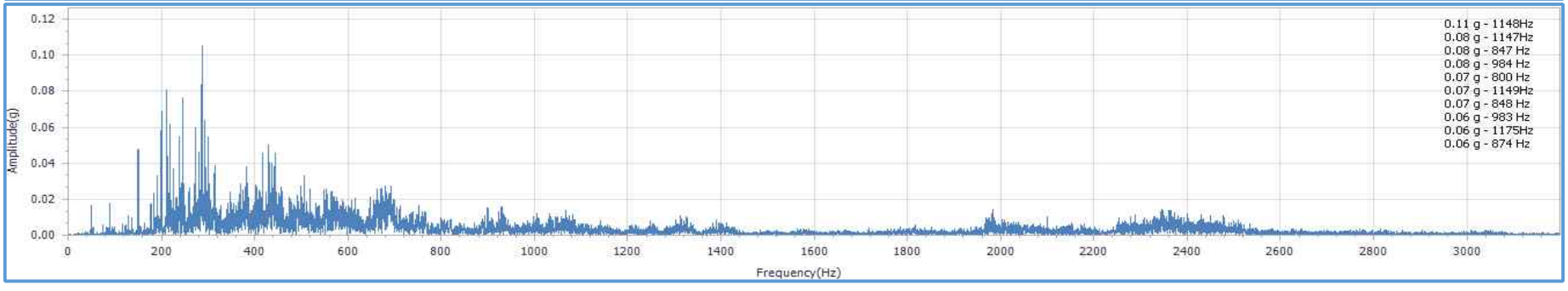


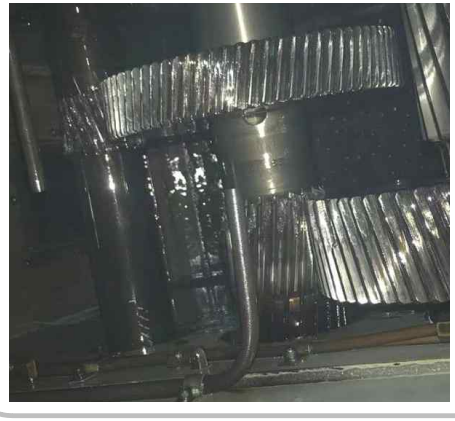


신호 처리 프로젝트 실습 6조

▶ 6조 감속기 1st/2nd Gear 손상

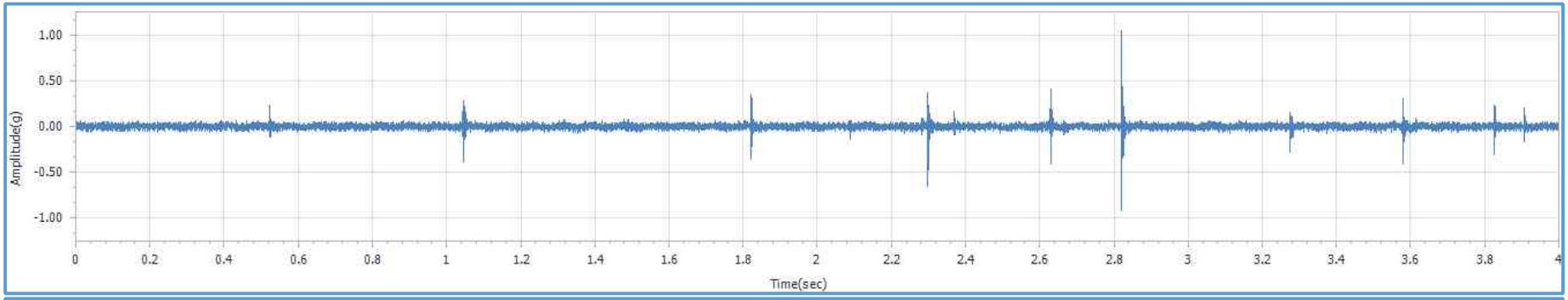


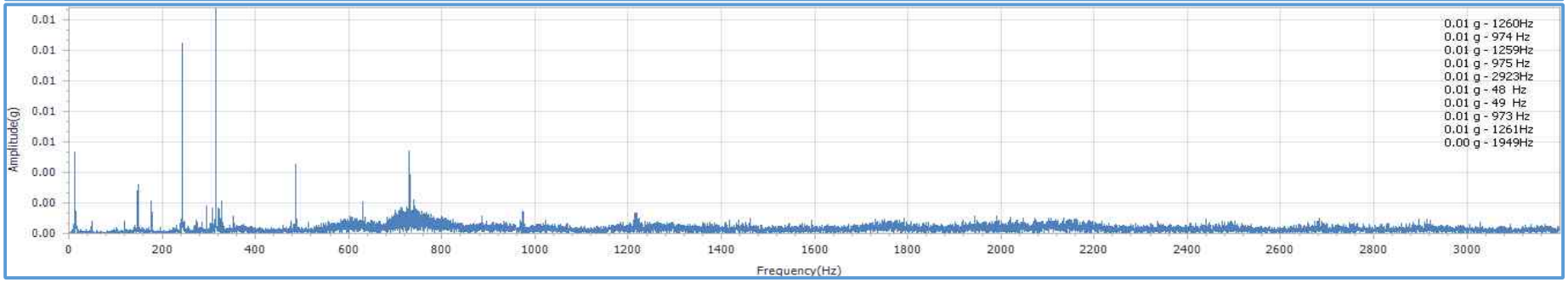




신호 처리 프로젝트 실습 1

▶ 7조 Feed Gear 손상

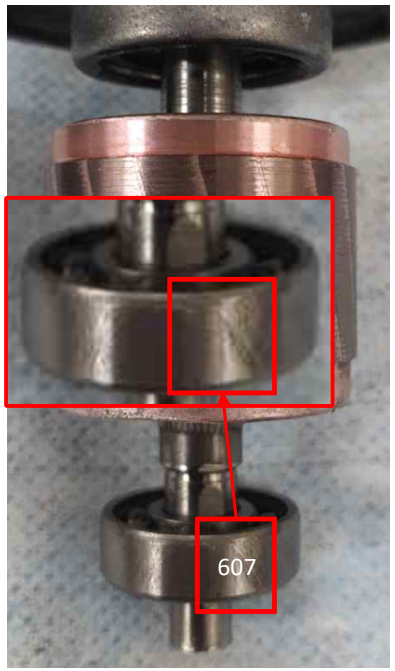




실습3_ 실습 장비 3w

◈ 실습 장비

Reducer 3rd Loose / Lubrication Analysis



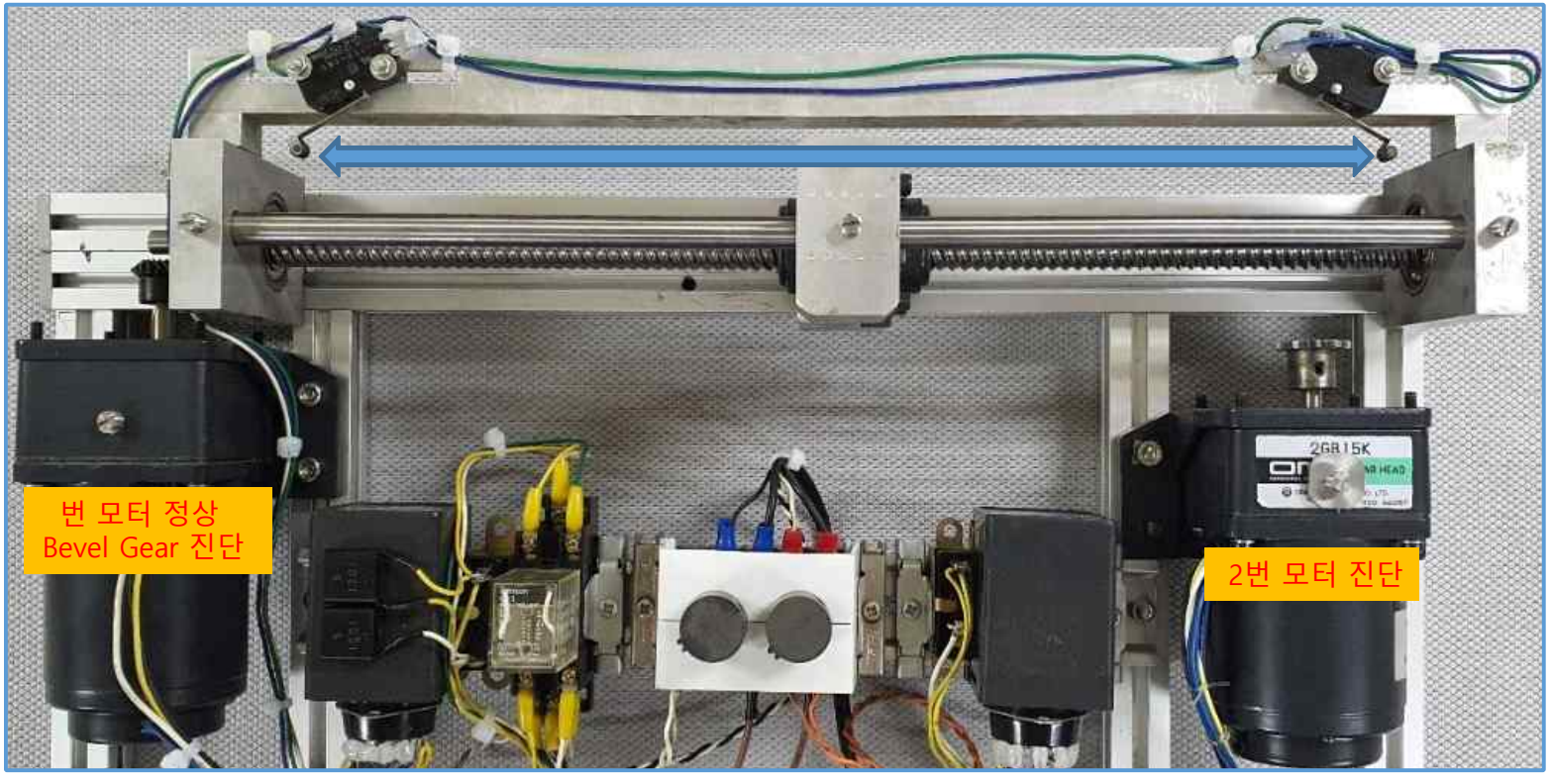
3w Motor Bearing Analysis

Bevel Gear Analysis

3w Motor 1st Gear Teeth Mesh Analysis

실습3_ 테스트 장비 1

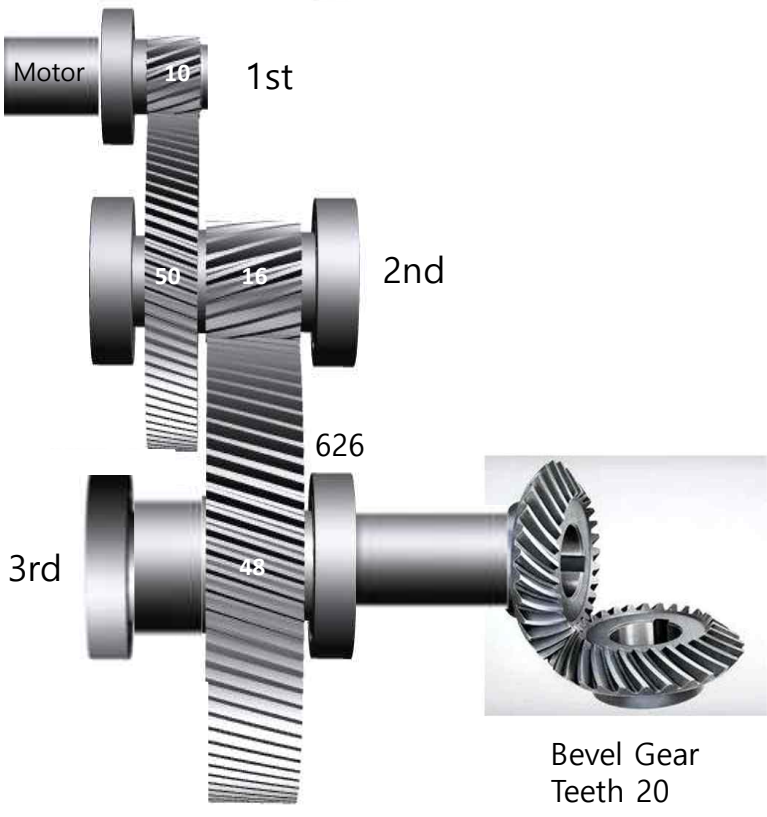
◈ 실습 장비 1



◈ 실습 장비 1 Motor 1780 rpm (2IJ3RGB-A)
1번 모터 베벨 기어 정 회전 역회전 맞물림 주파수 찾고 정회전 역회전 진동 특성에 대한 설명
2번 모터 모터 진동 분석

실습3_ 테스트 장비 1번 설비 정보

1. 설비 계략도



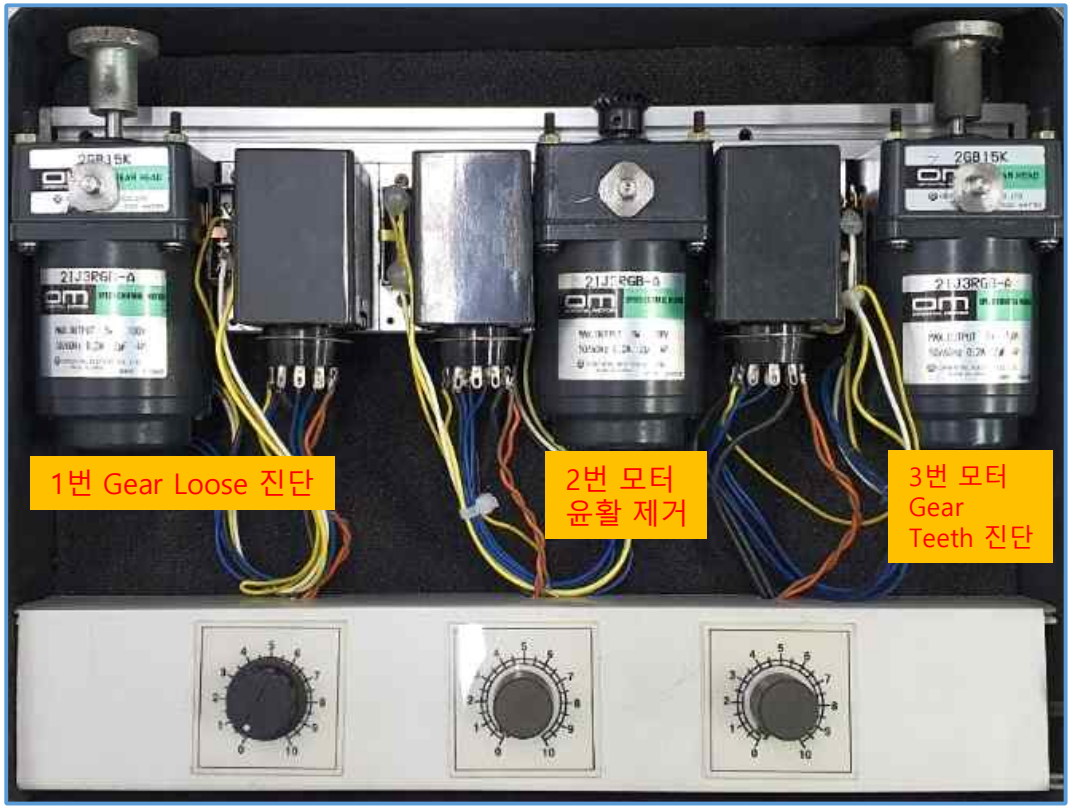
FAG Num	Description	Outer Race	Inner Race	2X Ball Spin	Train/Cage
607Z	Ball Bearing	2.550	4.450	3.414	0.365
626	Ball Bearing	1.723	4.442	3.445	0.365

Motor			Bearing	
Pole	rpm	kW	IB	OB
	1740	200	607Z	607Z

Bearing	Outer	Inner	Ball	Cage	GMF	Hz
607Z						
626Z						
1st	10					
2nd	50	16				
3rd		48	20			
Bevel			20			

실습4_ 테스트 장비 2번

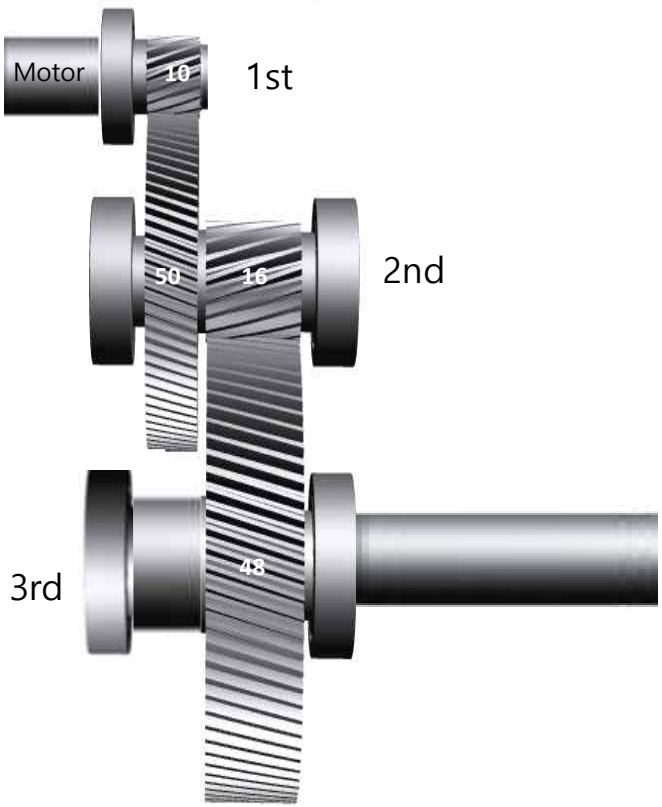
◈ 실습 장비 2



◈ 실습 장비 2
1번 모터 진동 원인 분석
2번 모터 윤활 문제 감속기 입니다.
3번 모터 진동 원인 분석

실습4_ 테스트 장비 2번 설비 정보

1. 설비 계략도



FAG Num	Description	Outer Race	Inner Race	2X Ball Spin	Train/Cage
607Z	Ball Bearing	2.550	4.450	3.414	0.365
626Z	Ball Bearing	2.558	4.442	3.445	0.365

Motor			**Bearing**	
Pole	**rpm**	**kW**	**IB**	**OB**
	1740(29Hz)	200	607	607

Bearing	Outer	Inner	Ball	Cage	GMF	Hz
모터 607Z						
3rd 626Z						
1st	10					
2nd	50	16				
3rd		48				